SmartPlant® Instrumentation

SPECSHEETS

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Implementation Team



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INTERGRAPH

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New SPI Spec Libraries

- Petrochemical Spec Library (SPI 2009 Library for SPI 2007.zip)
- Biopharma Spec Library (Biopharma_Spec_SPI 2007.zip)
- Additional SPI Specification Libraries and Forms
- SPI Specification Sheet Tips & Tricks

- **Petrochemical and Biopharma Spec Libraries**
  - These libraries are included with SPI 2009, however they must be restored for use and will need Infomaker 11 Service Pack 2.
  - These libraries will work in SPI 2007.6 and above. To install, use the Page Editor to open the Files and save as new Pages. Then create Forms from the Pages and use the SQL statement to create the required view.
  - To obtain these libraries – submit a service request to Intergraph

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SPI Petrochemical Spec Library

- This library is made up of 101 .psr files that need to be assembled in different ways to build complete spec forms.
- This Library is based on ISA S20.1 but not sanctioned by ISA.

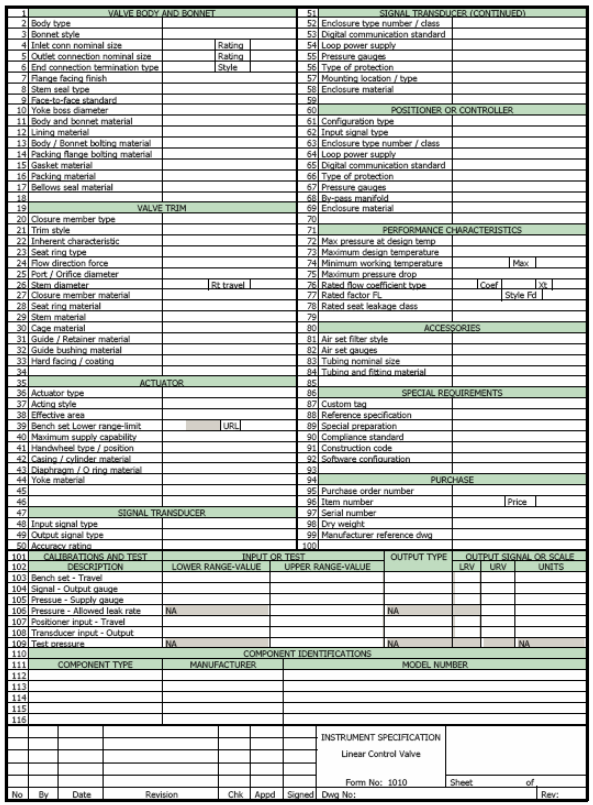
- Air-release air vacuum.psr
- Air-release valve.psr
- Analyzer device.psr
- Bimetalic Thermometer.psr
- Capacitance rf admit sw.psr
- Capacitance rf admit xmtr.psr
- Conductivity sensor.psr
- Conductivity-resist xmtr.psr
- Control valve aux2.psr
- Coriolis flow tube.psr
- Coriolis flow xmtr.psr
- Coriolis flowmeter assy.psr
- Diff press level xmtr-seal.psr
- Diff pressure device.psr
- Diff pressure gauge.psr
- Diff pressure level xmtr.psr
- Diff pressure switch.psr
- Diff pressure xmtr seal.psr
- Diff pressure xmtr.psr
- Disolved oxygen ozone.psr
- Displacer type xmtr.psr
- Elbow flowmeter.psr
- Electric actuator pg2.psr
- Electric actuator.psr
- Electromechanical limit Sw.psr
- Electronic weight xmtr.psr
- Entrained gas sensor.psr
- Entrained gas xmtr.psr
- Fieldbus junction.psr
- Filled-system temp sw.psr
- Filled-system temp xmtr.psr
- Filled-system thermometer.psr
- Flanged dp level xmtr.psr
- Float-displacer level sw.psr
- Flow device.psr
- Flow nozzle meter tube.psr
- Humidity dewpoint xmtr.psr
- Level device.psr
- Linear control valve.psr
- Linear MOV.psr
- Linear-rotary MOV pg2.psr
- Liquid level gage glass.psr
- Load cell transducer.psr
- Magnetic flow assembly.psr
- Magnetic flow tube.psr
- Magnetic flow xmtr.psr
- Magnetic liquid level gage.psr
- Magnetic liquid level sw.psr
- Multivariable Flow Xmtr.psr
- Non-contact ultrasonic.psr
- Nuclear level xmtr.psr
- Nuclear radiation level sw.psr
- Orifice plate1.psr
- Orifice plate assembly.psr
- Paddle flow switch.psr
- Paddle flow xmtr.psr
- PH orp sensor.psr
- pH-orp xmtr.psr
- Pilot pressure regulator.psr
- Pitot tube.psr
- Positive displacement xmtr.psr
- Positive displacement.psr
- Pressure device.psr
- Pressure gauge1.psr
- Pressure gauge Seal.psr
- Pressure regulator1.psr
- Pressure relief Valve1.psr
- Pressure switch1.psr
- Pressure xmtr seal.psr
- Pressure xmtr.psr
- Proximity sensor.psr
- Receiver gauge.psr
- Relief device.psr
- Resistance-tape level xmtr.psr
- Rotary control valve.psr
- Rotary level switch.psr
- Rotary MOV.psr
- RTD assembly thermowell.psr
- RTD tc temperature xmtr.psr
- Rupture disk assembly.psr
- Segmental wedge flowmeter.psr
- Sight flow indicator.psr
- Sonar flow sensor.psr
- Sonar flow Xmtr.psr
- Tank level gauge or Ind.psr
- Temperature device.psr
- Thermal mass flow switch.psr
- Thermal mass flowmeter.psr
- Thermal radiation temp.psr
- Thermocouple assembly.psr
- Thermowell protecting tube.psr
- Turbidity solids sensor.psr
- Turbidity Solids Xmtr.psr
- Ultrasonic contact level.psr
- Ultrasonic flow tube.psr
- Ultrasonic flow xmtr.psr
- Valve or reg device.psr
- V-cone flow element.psr
- Venturi or flow tube.psr
- Vortex swirl flowmeter.psr
- Weight Device.psr

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SPI Petrochemical Spec Library

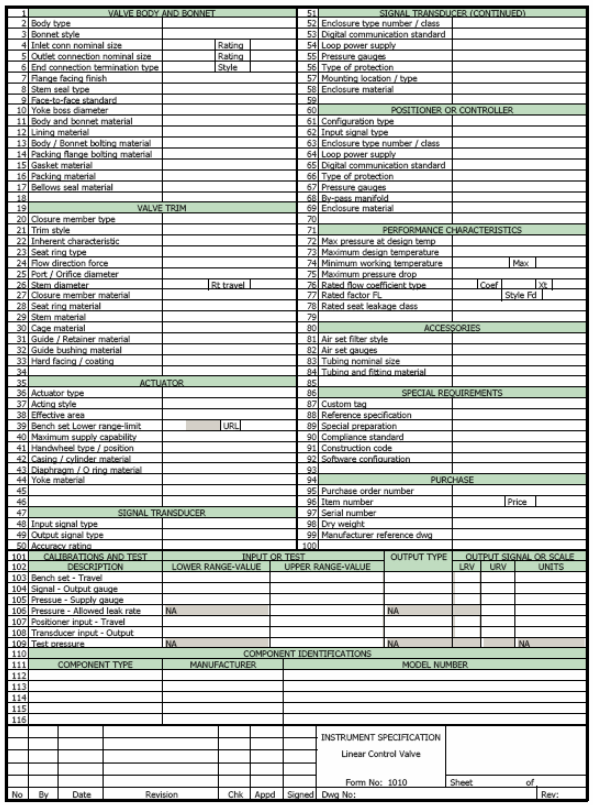
- An example of a Control Valve Spec Form will involve:



| | | |

SPI Petrochemical Spec Library

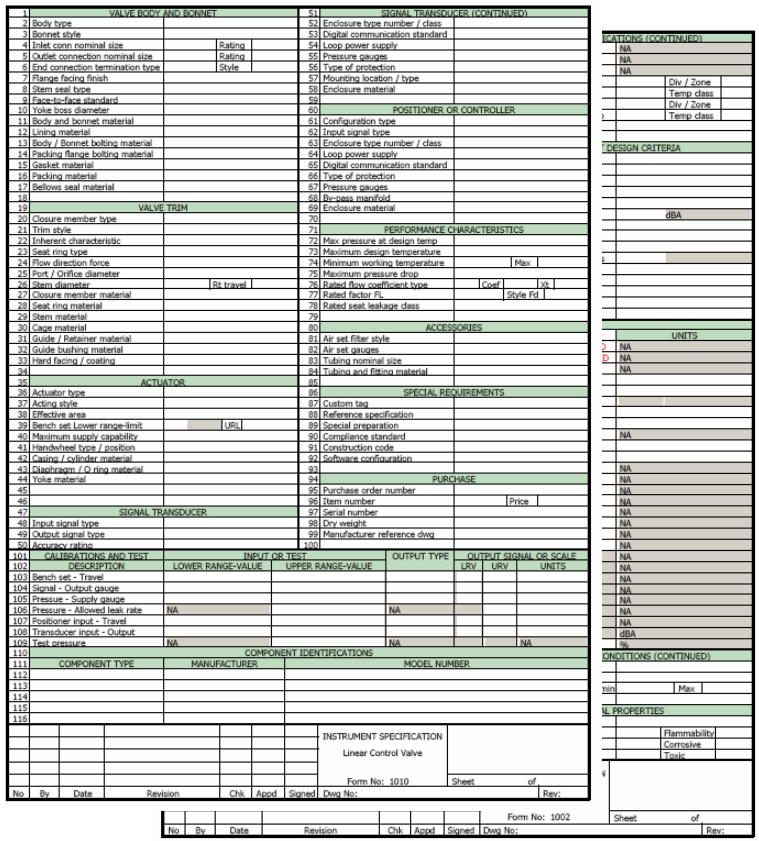
- Combine the Selected Valve Mechanics with the Process Spec



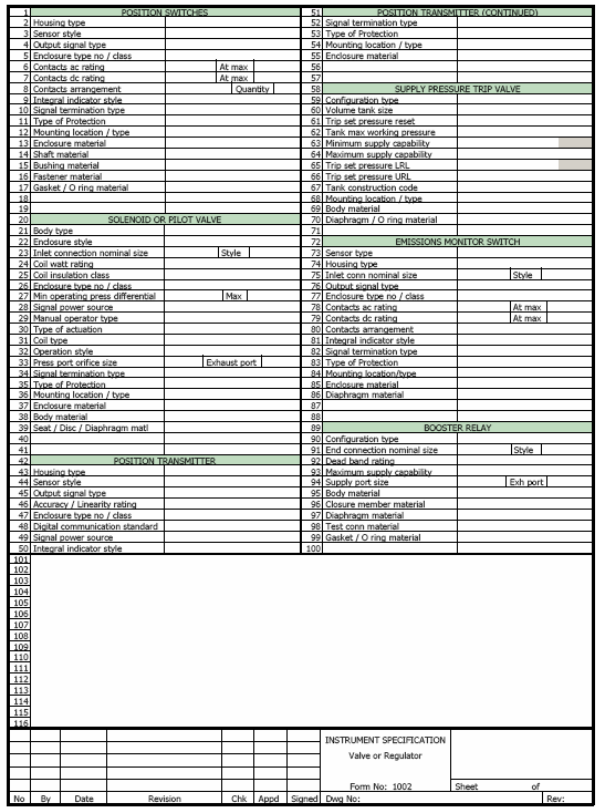
| | | |

SPI Petrochemical Spec Library

- An Auxiliary Device page is added to the Control Valve Spec



Valve + Design Conditions and



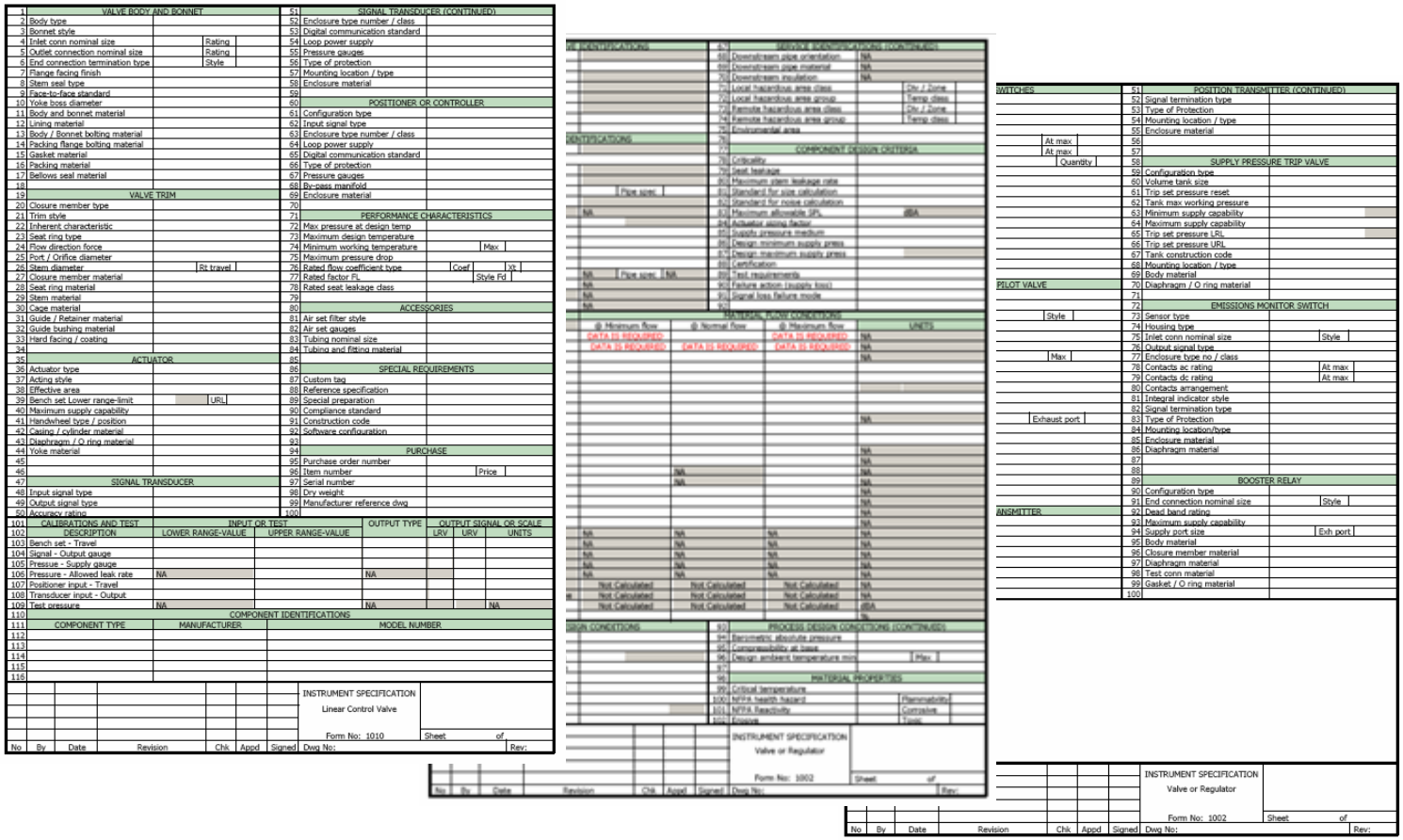
Valve auxiliary psr

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SPI Petrochemical Spec Library

- The resulting Spec sheet is three pages long and 330 data fields



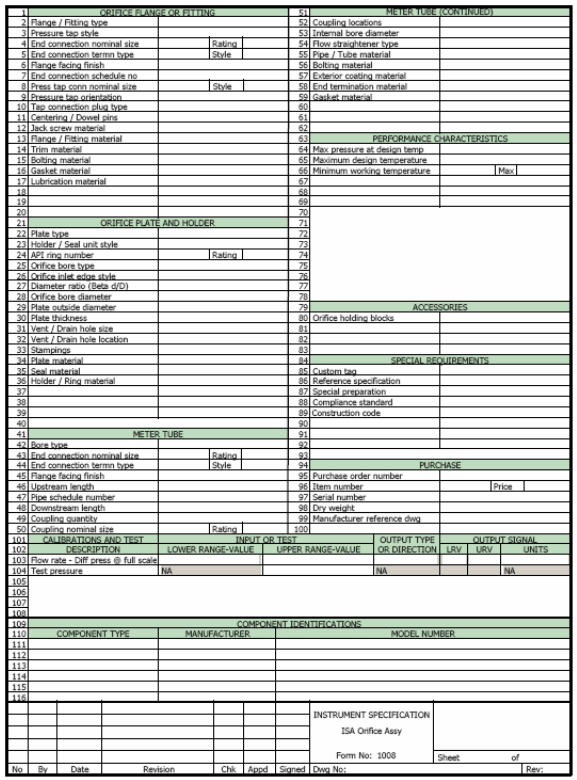
Complete Control Valve Form with three Pages

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SPI Petrochemical Spec Library

- An example of a Flow Element Form will involve:



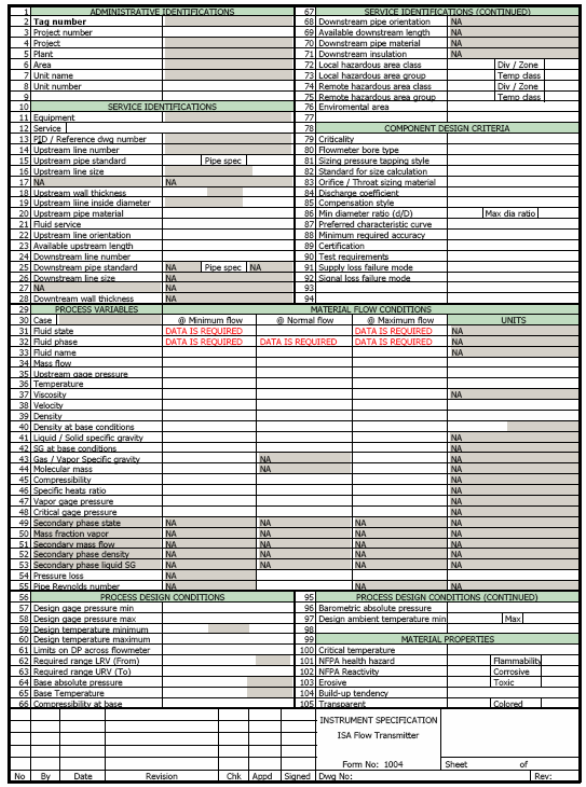
	ORIFICE FLANGE OR FITTING				METER TUBE (CONTINUED)		
2	Flange / Fitting type			52	Coupling locations		
3	Pressure tap style			53	Internal bore diameter		
4	End connection nominal size		Rating	54	Flow straightener type		
5	End connection termn type		Style	55	Pipe / Tube material		
6	Flange facing finish			56	Bolting material		
7	End connection schedule no			57	Exterior coating material		
8	Press tap conn nominal size		Style	58	End termination material		
9	Pressure tap orientation			59	Gasket material		
10	Tap connection plug type			60			
11	Centering / Dowel pins			61			
12	Jack screw material			62			
13	Flange / Fitting material			63	PERFORMANCE CHARACTERISTICS		
14	Trim material			64	Max pressure at design temp		
15	Bolting material			65	Maximum design temperature		
16	Gasket material			66	Minimum working temperature		Max
17	Lubrication material			67			
18				68			
19				69			
20				70			
21	ORIFICE PLATE AND HOLDER			71			
22	Plate type			72			
23	Holder / Seal unit style			73			
24	API ring number		Rating	74			
25	Orifice bore type			75			
26	Orifice inlet edge style			76			
27	Diameter ratio (Beta d/D)			77			
28	Orifice bore diameter			78			
29	Plate outside diameter			79	ACCESSORIES		
30	Plate thickness			80	Orifice holding blocks		
31	Vent / Drain hole size			81			
32	Vent / Drain hole location			82			
33	Stampings			83			
34	Plate material			84	SPECIAL REQUIREMENTS		
35	Seal material			85	Custom tag		
36	Holder / Ring material			86	Reference specification		
37				87	Special preparation		
38				88	Compliance standard		
39				89	Construction code		
40				90			
41	METER TUBE			91			
42	Bore type			92			
43	End connection nominal size		Rating	93			
44	End connection termn type		Style	94	PURCHASE		
45	Flange facing finish			95	Purchase order number		
46	Upstream length			96	Item number		Price
47	Pipe schedule number			97	Serial number		
48	Downstream length			98	Dry weight		
49	Coupling quantity			99	Manufacturer reference dwg		
50	Coupling nominal size		Rating	100			

101	CALIBRATIONS AND TEST	INPUT OR TEST		OUTPUT TYPE	OUTPUT SIGNAL		
102	DESCRIPTION	LOWER RANGE-VALUE	UPPER RANGE-VALUE	OR DIRECTION	LRV	URV	UNITS
103	Flow rate - Diff press @ full scale						
104	Test pressure	NA		NA			NA
105							
106							
107							
108							

109	COMPONENT IDENTIFICATIONS		
110	COMPONENT TYPE	MANUFACTURER	MODEL NUMBER
111			
112			
113			
114			
115			
116			

INSTRUMENT SPECIFICATION
ISA Orifice Assy
Form No: 1008
Sheet of

No	By	Date	Revision	Chk	Appd	Signed	Dwg No:	Rev:



	ADMINISTRATIVE IDENTIFICATIONS			SERVICE IDENTIFICATIONS (CONTINUED)		
2	**Tag number**		68	Downstream pipe orientation	NA	
3	Project number		69	Available downstream length	NA	
4	Project		70	Downstream pipe material	NA	
5	Plant		71	Downstream insulation	NA	
6	Area		72	Local hazardous area class		Div / Zone
7	Unit name		73	Local hazardous area group		Temp class
8	Unit number		74	Remote hazardous area class		Div / Zone
9			75	Remote hazardous area group		Temp class
10	SERVICE IDENTIFICATIONS		76	Enviromental area		
11	Equipment		77			
12	Service		78	COMPONENT DESIGN CRITERIA		
13	PID / Reference dwg number		79	Criticality		
14	Upstream line number		80	Flowmeter bore type		
15	Upstream pipe standard	Pipe spec	81	Sizing pressure tapping style		
16	Upstream line size		82	Standard for size calculation		
17	NA	NA	83	Orifice / Throat sizing material		
18	Upstream wall thickness		84	Discharge coefficient		
19	Upstream liine inside diameter		85	Compensation style		
20	Upstream pipe material		86	Min diameter ratio (d/D)		Max dia ratio
21	Fluid service		87	Preferred characteristic curve		
22	Upstream line orientation		88	Minimum required accuracy		
23	Available upstream length		89	Certification		
24	Downstream line number		90	Test requirements		
25	Downstream pipe standard	NA Pipe spec NA	91	Supply loss failure mode		
26	Downstream line size	NA	92	Signal loss failure mode		
27	NA	NA	93			
28	Downstream wall thickness	NA	94			

29	PROCESS VARIABLES	MATERIAL FLOW CONDITIONS			
30	Case	@ Minimum flow	@ Normal flow	@ Maximum flow	UNITS
31	Fluid state	DATA IS REQUIRED		DATA IS REQUIRED	NA
32	Fluid phase	DATA IS REQUIRED	DATA IS REQUIRED	DATA IS REQUIRED	NA
33	Fluid name				NA
34	Mass flow				
35	Upstream gage pressure				
36	Temperature				
37	Viscosity				NA
38	Velocity				
39	Density				
40	Density at base conditions				
41	Liquid / Solid specific gravity				NA
42	SG at base conditions				NA
43	Gas / Vapor Specific gravity		NA		NA
44	Molecular mass		NA		NA
45	Compressibility				NA
46	Specific heats ratio				NA
47	Vapor gage pressure				NA
48	Critical gage pressure				NA
49	Secondary phase state	NA	NA	NA	NA
50	Mass fraction vapor	NA	NA	NA	NA
51	Secondary mass flow	NA	NA	NA	NA
52	Secondary phase density	NA	NA	NA	NA
53	Secondary phase liquid SG	NA	NA	NA	NA
54	Pressure loss	NA			
55	Pipe Reynolds number	NA		NA	NA

56	PROCESS DESIGN CONDITIONS		95	PROCESS DESIGN CONDITIONS (CONTINUED)		
57	Design gage pressure min		96	Barometric absolute pressure		
58	Design gage pressure max		97	Design ambient temperature min		Max
59	Design temperature minimum		98			
60	Design temperature maximum		99	MATERIAL PROPERTIES		
61	Limits on DP across flowmeter		100	Critical temperature		
62	Required range LRV (From)		101	NFPA health hazard		Flammability
63	Required range URV (To)		102	NFPA Reactivity		Corrosive
64	Base absolute pressure		103	Erosive		Toxic
65	Base Temperature		104	Build-up tendency		
66	Compressibility at base		105	Transparent		Colored

INSTRUMENT SPECIFICATION
ISA Flow Transmitter
Form No: 1004
Sheet of

No	By	Date	Revision	Chk	Appd	Signed	Dwg No:	Rev:

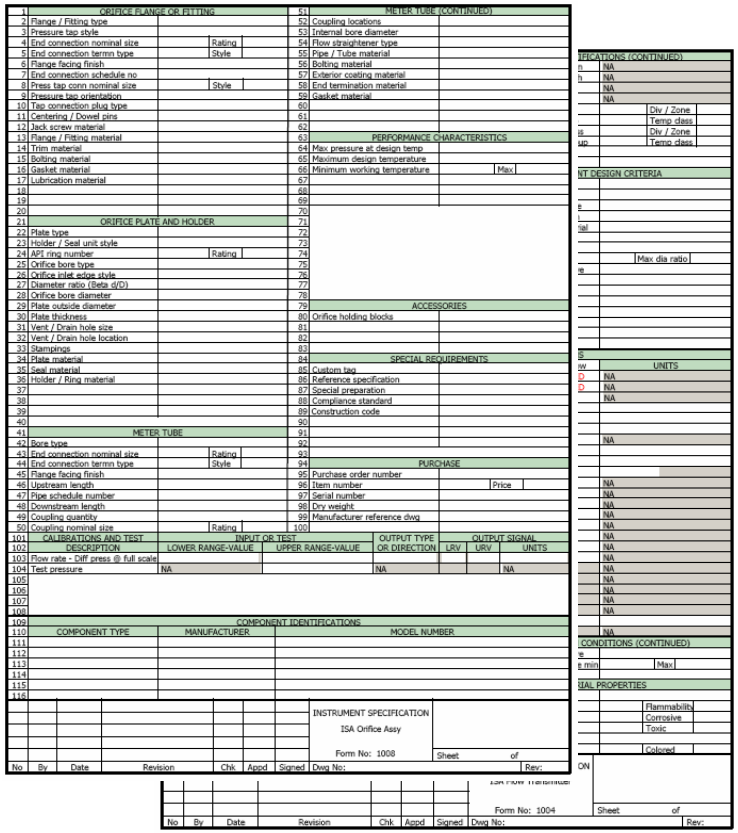
An Orifice Plate Assembly page and a Flow Process conditions page

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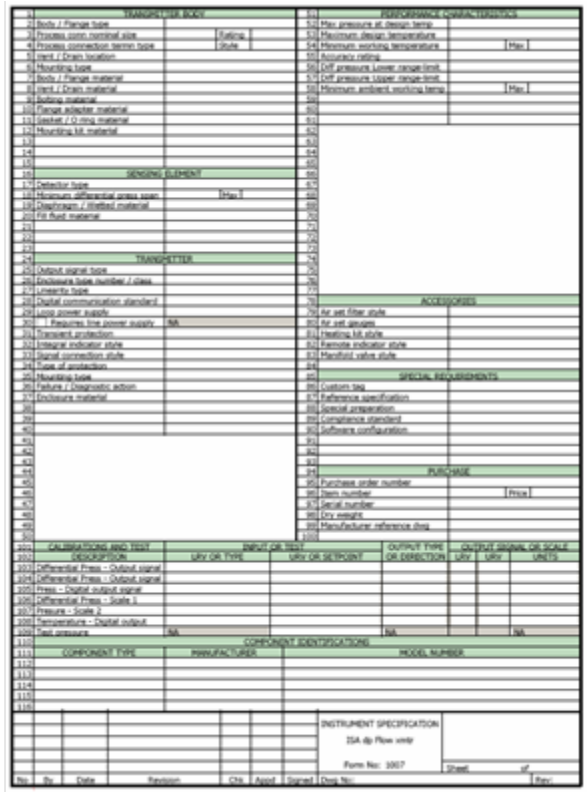
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SPI Petrochemical Spec Library

- A complete Flow element and transmitter will require:



The Flow Element Form with a



Pressure Transmitter Form

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SPI Petrochemical Spec Library

- **Issues with SPI Petrochemical Spec Library**
  - While the ISO9000 Title Block is nice the Notes Page reverts to Out of the Box SPI Title Block
  - Component Identification Areas are only associated to the database for one of the Component Types and do not contain the instrument Tag Numbers

Page Title

110	COMPONENT IDENTIFICATIONS							
111	COMPONENT TYPE			MANUFACTURER		MODEL NUMBER		
112	spec_cmpnt_type			cmpnt_mfr_id		cmpnt_mod_id		
113	spec_udf_c410			spec_udf_c68		spec_udf_c67		
114	spec_udf_c411			spec_udf_c57		spec_udf_c56		
115	spec_udf_c412			spec_udf_c311		spec_udf_c415		
116	spec_udf_c413			spec_udf_c27		spec_udf_c07		
							INSTRUMENT SPECIFICATION ISA dp Flow xmtr Form No: 1007	bitmap (ProfileString ("intools.ini", Sheet spec_she of spec_she
No	By	Date	Revision	Chk	Appd	Signed	Dwg No: dwg_name	Rev:

Notes Title

				INSTRUMENT SPECIFICATION Control Valve		
					Sheet	of
No.	By	Date	Revision	Code:101	Dwg. No.: SSD-21 -FV -1A11	Rev.:

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SPI Petrochemical Spec Library

- **Issues with SPI Petrochemical Spec Library**
  - The Tag Number and P&ID do not appear on every page
  - The Forms are very complex to build and associate the proper pages to complete the required data
  - The page descriptions should give Form relationships and Process Function
  - Some Pages have wasted space
  - The resulting Forms are too complex and time consuming to fill out properly or transmit to vendors for sizing and selection
  - The pages contain “N/A” or Non-functional fields with no value
  - Blank lines with Header and Data areas do not allow data input

22	Upstream line orientation	spec_udf_c442		
23	Available upstream length	spec_udf_rspec_udf_		
24	Downstream line number	pd_udf_c01		
25	Downstream pipe standard	NA	Pipe spec	NA
26	Downstream line size	NA		
27	NA	NA		

29	PROCESS VARIABLES		MATERIAL
30	Case case_id	spec_udf_c725	spec_udf_c726
31	Fluid state	spec_udf_c760	pd_fluid_phase
32	Fluid phase	spec_udf_c762	pd_fluid_stat
33	Fluid name	spec_udf_c770	pd_fluid_name
34	if(isNull (pd_flow_uflo) or pd_flo	pd_flow_min	pd_flow_nor

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SPI Pharmaceutical Spec Library

- This library is made up of 74 .psr pages that are stand alone forms. Features many new pages for specialty instruments
- Per CR#77777 ID#91 was implemented in SPI as add-in library

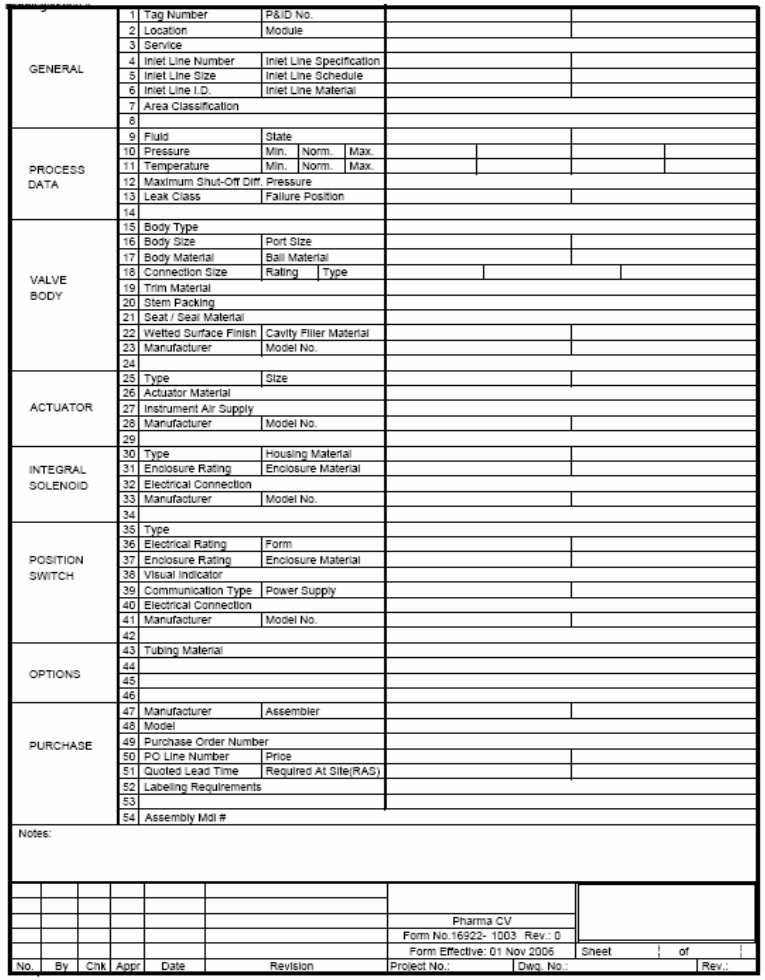


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SPI Pharmaceutical Spec Library

- Valve Spec Selections Include:
  - 7101 - Temperature Regulator
  - 7102 - Pressure Flow Regulator
  - 7103 - Blanketing Regulator
  - 7201 - Ball Valve
  - 7202 - 4 Way Valve
  - 7203 - 3 Way Valve
  - 7204 - Diaphragm Valve
  - 7205 - Butterfly Valve On Off
  - 7206 - 3 Way Diverter Valve
  - 7207 - Sanitary Multi-port Valve
  - 7301 - Globe Valve
  - 7302 - Sanitary Control Valve
  - 7303 - Butterfly Control Valve
  - 7304 - V Ball Valve



GENERAL	1	Tag Number	P&ID No.								
	2	Location	Module								
	3	Service									
	4	Inlet Line Number	Inlet Line Specification								
	5	Inlet Line Size	Inlet Line Schedule								
	6	Inlet Line I.D.	Inlet Line Material								
	7	Area Classification									
	8										
PROCESS DATA	9	Fluid	State								
	10	Pressure	Min.	Norm.	Max.						
	11	Temperature	Min.	Norm.	Max.						
	12	Maximum Shut-Off Diff. Pressure									
	13	Leak Class	Failure Position								
	14										
VALVE BODY	15	Body Type									
	16	Body Size	Port Size								
	17	Body Material	Ball Material								
	18	Connection Size	Rating	Type							
	19	Trim Material									
	20	Stem Packing									
	21	Seat / Seal Material									
	22	Wetted Surface Finish	Cavity Filler Material								
	23	Manufacturer	Model No.								
	24										
ACTUATOR	25	Type	Size								
	26	Actuator Material									
	27	Instrument Air Supply									
	28	Manufacturer	Model No.								
	29										
INTEGRAL SOLENOID	30	Type	Housing Material								
	31	Enclosure Rating	Enclosure Material								
	32	Electrical Connection									
	33	Manufacturer	Model No.								
	34										
POSITION SWITCH	35	Type									
	36	Electrical Rating	Form								
	37	Enclosure Rating	Enclosure Material								
	38	Visual Indicator									
	39	Communication Type	Power Supply								
	40	Electrical Connection									
	41	Manufacturer	Model No.								
	42										
OPTIONS	43	Tubing Material									
	44										
	45										
	46										
PURCHASE	47	Manufacturer	Assembler								
	48	Model									
	49	Purchase Order Number									
	50	PO Line Number	Price								
	51	Quoted Lead Time	Required At Site(RAS)								
	52	Labeling Requirements									
	53										
	54	Assembly Mdl #									

Notes:

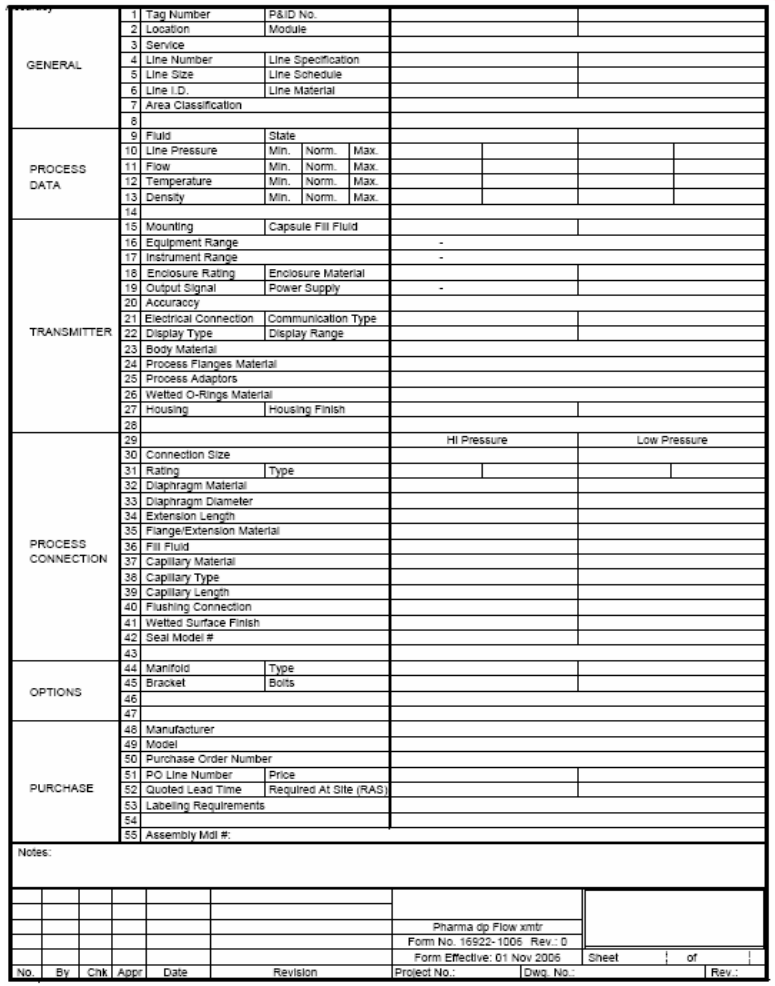
No.	By	Chk	Appr	Date	Revision	Pharma CV Form No.16922- 1003 Rev.: 0 Form Effective: 01 Nov 2006	Sheet of
						Project No.: Dwg. No.:	Rev.:

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SPI Pharmaceutical Spec Library

- Flowmeter Spec Selections include:
  - 2501 - Coriolis Mass Flowmeter
  - 2502 - Magnetic Flowmeter
  - 2503 - Thermal Mass Flowmeter
  - 2504 - Turbine Flowmeter
  - 2505 - DP Transmitter flow
  - 2506 - Positive Displacement Flowmeter
  - 2507 - Vortex Flowmeter
  - 2508 - Ultrasonic Flowmeter
- Flow Element Selections include:
  - 1101 Orifice Plate
  - 1102 Annubar
  - 1103 Rotameter
  - 1104 Venturi Tube



GENERAL	1	Tag Number	P&ID No.							
	2	Location	Module							
	3	Service								
	4	Line Number	Line Specification							
	5	Line Size	Line Schedule							
	6	Line I.D.	Line Material							
	7	Area Classification								
	8									
PROCESS DATA	9	Fluid	State							
	10	Line Pressure	Min.	Norm.	Max.					
	11	Flow	Min.	Norm.	Max.					
	12	Temperature	Min.	Norm.	Max.					
	13	Density	Min.	Norm.	Max.					
	14									
TRANSMITTER	15	Mounting	Capsule Fill Fluid							
	16	Equipment Range				-				
	17	Instrument Range				-				
	18	Enclosure Rating	Enclosure Material							
	19	Output Signal	Power Supply			-				
	20	Accuracy								
	21	Electrical Connection	Communication Type							
	22	Display Type	Display Range							
	23	Body Material								
	24	Process Flanges Material								
	25	Process Adaptors								
	26	Wetted O-Rings Material								
	27	Housing	Housing Finish							
	28									
PROCESS CONNECTION	29					HI Pressure		Low Pressure		
	30	Connection Size								
	31	Rating	Type							
	32	Diaphragm Material								
	33	Diaphragm Diameter								
	34	Extension Length								
	35	Flange/Extension Material								
	36	Fill Fluid								
	37	Capillary Material								
	38	Capillary Type								
	39	Capillary Length								
	40	Flushing Connection								
	41	Wetted Surface Finish								
	42	Seal Model #								
	43									
OPTIONS	44	Manifold	Type							
	45	Bracket	Bolts							
	46									
	47									
PURCHASE	48	Manufacturer								
	49	Model								
	50	Purchase Order Number								
	51	PO Line Number	Price							
	52	Quoted Lead Time	Required At Site (RAS)							
	53	Labeling Requirements								
	54									
	55	Assembly Mdl #:								

Notes:

No.	By	Chk	Appr	Date	Revision

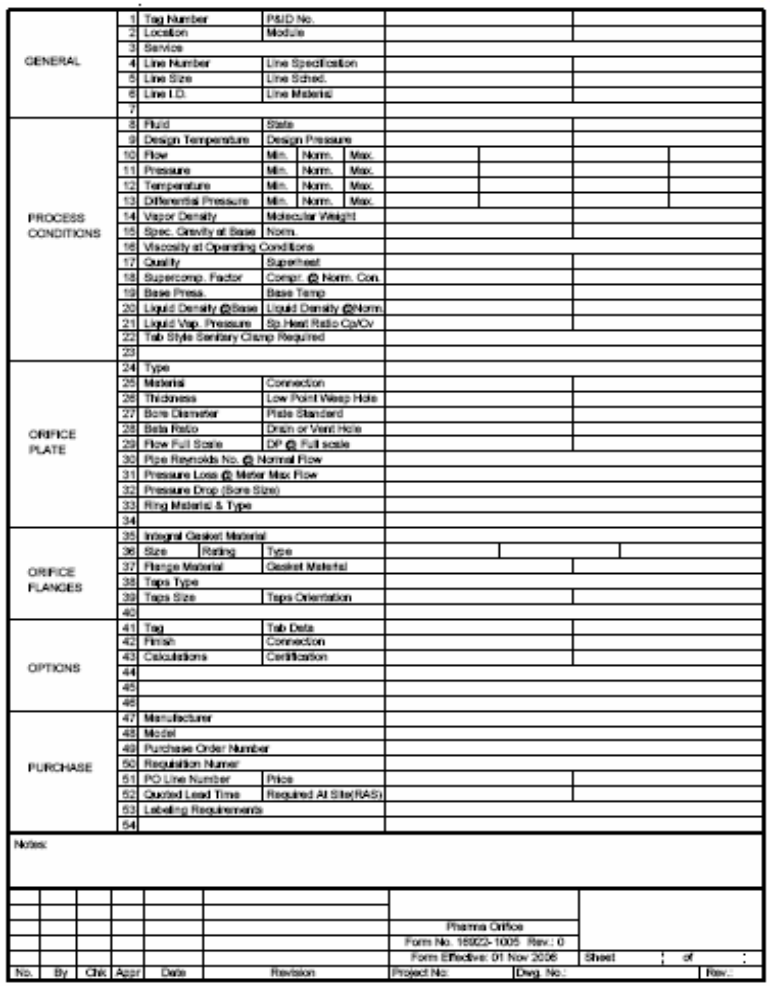
Pharma dp Flow xmtr
Form No. 16922-1006 Rev.: 0
Form Effective: 01 Nov 2005
Sheet of
Project No.: Dwg. No.: Rev.:

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SPI Pharmaceutical Spec Library

- Specialty Spec Selections include:
  - 2610 - Total Organic Carbon Analyzer
  - 2609 - Optical Density Analyzer
  - 2611 - Turbidity Analyzer
  - 2701 - Weigh Systems, Load Cell
  - 2702 - Weigh Bench and Floor Scales
  - 2703 - Weigh Precision Scales
  - 4105 - Proximity for Bubble Trap
  - 5101 - Bi-metallic Temp Indicator
  - 7402 - Conservation Vent
  - 7401 - Rupture Disk
  - 7404 - Flame Arrestor
  - 9305 - Speed Transmitter



GENERAL	1	Tag Number	P&ID No.								
	2	Location	Module								
	3	Service									
	4	Line Number	Line Specification								
	5	Line Size	Line Sched.								
	6	Line I.D.	Line Material								
	7										
PROCESS CONDITIONS	8	Fluid	State								
	9	Design Temperature	Design Pressure								
	10	Flow	Min.	Norm.	Max.						
	11	Pressure	Min.	Norm.	Max.						
	12	Temperature	Min.	Norm.	Max.						
	13	Differential Pressure	Min.	Norm.	Max.						
	14	Vapor Density	Molecular Weight								
	15	Spec. Gravity at Base	Norm.								
	16	Viscosity at Operating Conditions									
	17	Quality	Superheat								
	18	Supercomp. Factor	Compr. @ Norm. Con.								
	19	Base Press.	Base Temp								
	20	Liquid Density @Base	Liquid Density @Norm								
	21	Liquid Vap. Pressure	Sp.Heat Ratio Cp/Cv								
	22	Tab Style Sanitary Clamp Required									
	23										
ORIFICE PLATE	24	Type									
	25	Material	Connection								
	26	Thickness	Low Point Weep Hole								
	27	Bore Diameter	Plate Standard								
	28	Beta Ratio	Drain or Vent Hole								
	29	Flow Full Scale	DP @ Full scale								
	30	Pipe Reynolds No. @ Normal Flow									
	31	Pressure Loss @ Meter Max Flow									
	32	Pressure Drop (Bore Size)									
	33	Ring Material & Type									
	34										
ORIFICE FLANGES	35	Integral Gasket Material									
	36	Size	Rating	Type							
	37	Flange Material	Gasket Material								
	38	Taps Type									
	39	Taps Size	Taps Orientation								
	40										
OPTIONS	41	Tag	Tab Data								
	42	Finish	Connection								
	43	Calculations	Certification								
	44										
	45										
	46										
PURCHASE	47	Manufacturer									
	48	Model									
	49	Purchase Order Number									
	50	Requisition Number									
	51	PO Line Number	Price								
	52	Quoted Lead Time	Required At Site(RAS)								
	53	Labeling Requirements									
	54										

Notes:

No.	By	Chk	Appr	Date	Revision			
						Pharma Office		
						Form No. 16923-1005 Rev.: 0		
						Form Effective: 01 Nov 2006	Sheet : of :	
						Project No:	Dwg. No.:	Rev.:

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SPI Pharmaceutical Spec Library

- **Pros of SPI 2007 Pharmaceutical Spec Library**
  - ISO9000 Title Block blends with SPI Out of the Box but Notes Page reverts to Out of the Box Title Block
  - P&ID Number on every Spec Page
  - Has a selection of specialty Spec forms that supplement the SPI Out of the Box Specs
  - The library matches the ISA20 Excel Library for Pharmaceuticals



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Out of the Box Spec Libraries for SPI

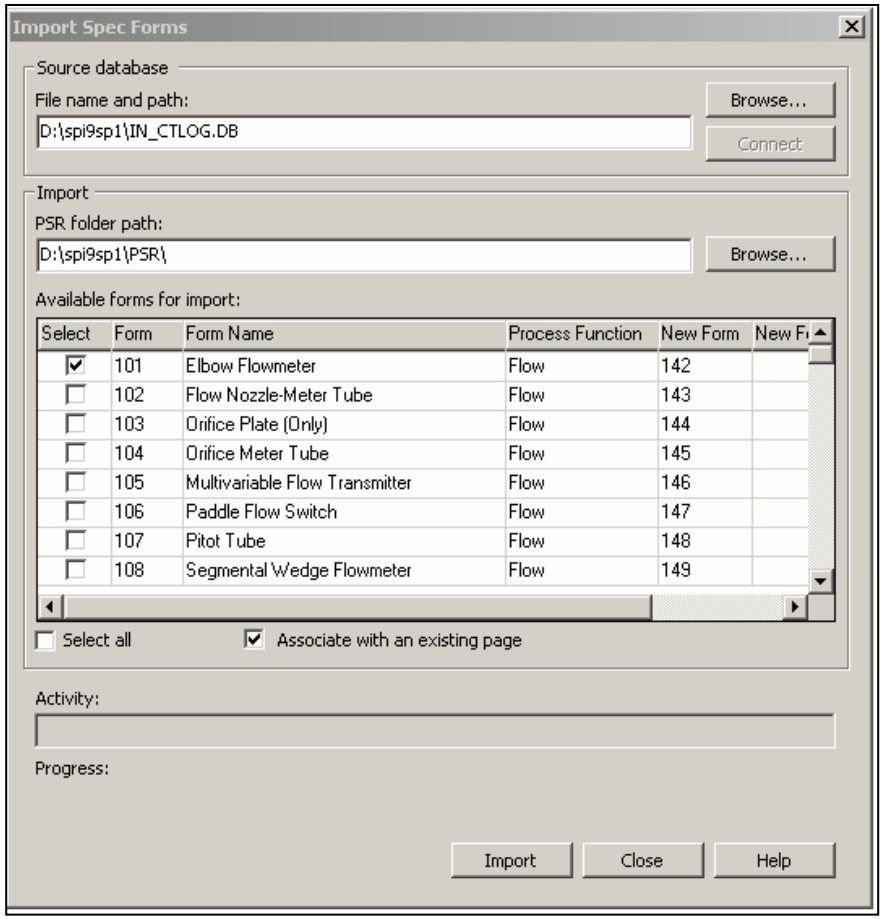
- **SPI Out of the Box Specification Library Pros**
  - Complete Library for most common Instruments
  - Easily modified in SPI to fit most project specific needs
  - Handles Single, Multi-Tag and Multi Item Specs
  - Most SPI users are trained and skilled in the SPI Spec Library
- **SPI Out of the Box Specification Library Cons**
  - Title Box is not ISO-9000 compliant (no check or approved)
  - P&ID number not on all Spec forms
  - Forms have not been updated in several SPI versions
  - Requires modifications to Specs on almost every project
- **Houston SPI Spec Committee suggested library update in 2004**
  - Add ISO-9000 Title Box to all Out of the Box Spec Forms
  - Add P&ID Number to all Out of the Box Spec Forms among others...
  - Reintroduced to Intergraph on February10, 2009 (see spi-ltuf.org archive)

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Other Spec Libraries for SPI

- **Add-In Additional Spec Forms** loaded from Admin Module



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Other Spec Libraries for SPI

- **Add-In Additional Spec Forms** – 90 additional Spec Forms

101	Elbow Flowmeter	214	Nuclear Level Transmitter	519	Disolved O2 / Ozone / Cl2 Xmtr
102	Flow Nozzle-Meter Tube	215	Resistance-Tape Level Xmtr	534	pH-ORP Sensor
103	Orifice Plate (Only)	217	Non-contact Ultrasonic Xmtr	535	pH-ORP Transmitter
104	Orifice Meter Tube	230	Capacitance RF Admit Switch	537	Humidity-Dewpoint Xmtr
105	Multivariable Flow Transmitter	231	Float or Displacer Level Sw	538	Conductivity Sensor
106	Paddle Flow Switch	232	Nuclear Radiation Level Sw	539	Entrained/Holdup Gas Sensor
107	Pitot Tube	235	Ultrasonic Contact Level Sw	540	Turbidity-Solids Sensor
108	Segmental Wedge Flowmeter	237	Rotary Level Switch	541	Turbidity-Solids Xmtr
109	Paddle Flow Transmitter	238	Tank Level Gauge or Ind	542	Entrained/Holdup Gas Xmtr
110	V-Cone Flow Element	250	Liquid Level Gage Glass	550	Conductivity-Resistivity Xmtr
111	Venturi-Flow Tube	251	Magnetic Liquid Level Gage	600	Linear Motion Control Valve
132	Magnetic Flowmeter Assembly	252	Magnetic Liquid Level Switch	601	Rotary Motion Control Valve
133	Magnetic Flowmeter Tube	253	Diff Press Level Xmtr w/Seal	602	Linear Valve w/Auxiliaries
134	Ultrasonic Flowmeter Tube	300	Pressure Gauge	604	Linear Motion MOV
135	Thermal Mass Flowmeter	301	Pressure Gauge w/Seal	605	Rotary Motion MOV
136	Thermal Mass Flow Switch	310	Diff Pressure Gauge	606	Electric Actuator
137	Magnetic Flowmeter Transmitter	320	Pressure Transmitter	607	Rotary Valve w/Auxilaries
138	Vortex-Swirl Flowmeter	321	Pressure Transmitter w/Seal	611	Pressure Regulator
139	Sonar Flowmeter Sensor	330	Diff Pressure Transmitter	614	Air-Release Air-Vacuum Valve
141	Positive Displacement Ind	331	Diff Pressure Xmtr w/Seal	615	Air Release Valve
142	Positive Displacement Xmtr	340	Pressure Switch	616	Pilot Operated Regulator
143	Sonar Flowmeter Transmitter	350	Diff Pressure Switch	790	Rupture Disk Assembly
152	Coriolis Flowmeter Assembly	400	Bimetallic Thermoter	792	Pressure Relief Valve
153	Coriolis Flowmeter Tube	410	Filled-System Thermometer	802	Receiver Gauge
154	Coriolis Flow Transmitter	411	Filled-System Temp Switch	821	Load Cell Transducer
155	Ultrasonic Flowmeter Xmtr	412	Filled-System Temp Xmtr	822	Electronic Weight Transmitter
161	Sight Flow Indicator	420	RTD Assembly-Thermowell	851	Fieldbus Junction Box
210	Capacitance-RF Admit Xmtr	422	RTD / TC temperature Xmtr	861	Electromechanical Limit Sw
211	Flange Mount DP Level Xmtr	430	Thermocouple Assembly	862	Proximity Sensor
212	Displacer Type Level Xmtr	440	Thermal Radiation Transmitter		
213	Diff Pressure Level Xmtr	450	Thermowell / Protecting Tube		

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Other Spec Libraries for SPI

- **The Overload Services Inc. SPI Spec Library** is made up of 135 Forms and is the most comprehensive spec Library for SPI

100	Analyzer	300	Annubar	406	Junction Box	515	Level Instrument Nuclear
101	Combustible Gas Toxic Detector	301	Averaging Pitot Tube	407	Instrument Cable	600	DP Transmitter
102	Conductivity Element	302	Flow DP Transmitter	408	Instrinsically Safe Barrier	601	DP Transmitter Smart
103	Conduct Transmitter Controller	303	Flow DP Transmitter Smart	409	Limit Switch	602	Pressure Controller Pneumatic
104	Consistency Meter	304	Flow Nozzle	410	Vibration Switch	603	Pressure Gauge
105	Density Transmitter	305	Flow Switch	411	Nuclear Instrument	604	Pressure Switch
106	Gas Density Transmitter	306	Flowmeter Magnetic	412	Power Supply	605	Pressure Transmitter Smart
107	Hydrogen Sulfide Monitor	307	Flowmeter Mass	413	Receiver Controller	606	Pressure Transmitter
108	Oxygen Analyzer	308	Flowmeter Posi Displacement	414	Transducer	607	Local Electronic Indicator
109	pH Element	309	Flowmeter Thermal Mass	415	Traps and Drainers	608	Miscellaneious Pressure
110	pH Transmitter Controller	310	Flowmeter Turbine	416	Vibration Monitor	700	Potentiometer Instrument
111	Process Analyzer	311	Flowmeter Vortex	417	Miscellaneous	701	Temp Controller Pneumatic
112	Complex Analyzer	312	Rotameter	419	Speed Instrument	702	Temp Electronic Transmitter
113	Miscellaneous Analyzer	313	Orifice Meter Run	421	Apparatus Barrier	703	Temperature Element
114	Analyzer Equipment	314	Orifice Plate	422	General Function Req.	704	Temperature Gauge
115	Analyzer Stream Sheet 1	315	Orifice Plate Style 2	423	Hook Up Item	705	Temp Instrument Filled System
116	Analyzer Stream Sheet 2	316	Restriction Orifice	424	Instrument Cable	706	Temperature Transmitter Smart
200	On/Off Valve	317	Restriction Orifice Style 2	425	Loop	707	Thermowell
201	Butterfly Valve	318	Venturi Tube	426	Panel (Junction Box)	708	Temperature Switch
202	Control Valve	319	Segmental Wedge	500	Interface Detector	800	Conservation Vent
203	Control Valve Style 2	320	Integral Orifice	501	Level DP Transmitter	801	Pressure Relief Valve
204	DP Regulator	321	Integral Orifice Style 2	502	Level DP Transmitter Smart	802	Rupture Disk
205	Emergency Block Valve	322	Wedge Flowmeter	503	Level Displacement Instrument	803	Releif Valve Style 2
206	Globe Valve Special	323	Ultrasonic Flow Instr	504	Level Electronic Transmitter	804	Miscellaneous Relief Valve
207	Solenoid Valve	324	Miscellaneous Flow	505	Level Gauge	806	Profiler
208	Press Control Valve Self Act	325	V-Cone Flowmeter	506	Level Instrument Capacitance	807	Nuclear Level Profiler
209	Pressure Regulator	326	Flow Function Req.	507	Level Transmitter Flange Mount	810	Relief Valve
210	Temperature Regulator	399	Inline Diffuser	508	Oil Water Monitor and Probe	900	Combustible Gas Detector
211	Motor Operated Valve	400	Annunciator	509	Ultrasonic Level Transmitter	901	Optical Flame Detector
212	Miscellaneous Control Valve	401	Annunciator Style 2	510	Magnetic Level Gauge	902	Smoke Detector
213	Control Valve Hybrid Case	402	Beacon	511	Radar Level Transmitter	903	Heat Detector or Fusible Link
214	Control Valve Function Req.	403	Burner Management	512	Miscellaneous Level	904	Manual Station
298	Electric Operated Check Valve	404	Condensing Seal and Drip Pots	513	Level Instr Guided Wave Radar	905	Alarm Horn
299	Motor Operated Valve	405	Fire Detector and Monitor	514	Level Switch	906	Analyzer Indicator Switch

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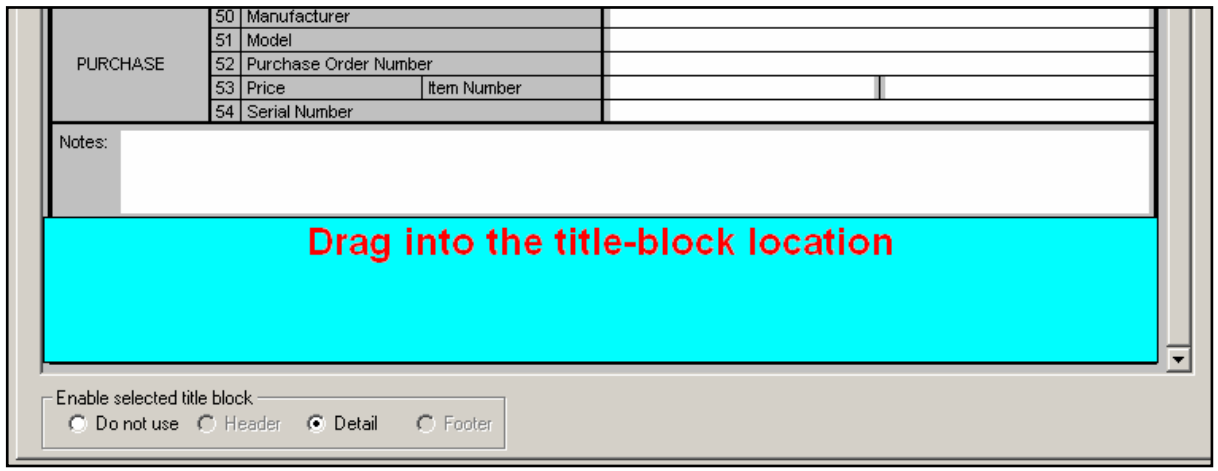
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Other Spec Libraries for SPI

- **Overload Services Inc. SPI Spec Library Pros**
  - Comes with OSI Seed file which is used by several Owner Operators
  - P&ID Number on every Spec Page
  - Title Block is ISO-9000 compliant but Notes Page reverts to Out of the Box Title Block
  - Additional Specs available for Consolidated Relief Valves and Level Transmitter Calculations
- **Overload Services Inc. SPI Spec Library Cons**
  - Must be purchased from Overload Services Inc. as a standalone library or as part of a OSI SPI Seed file
- Overload Services has also developed an ISA SPI Spec Library similar to the Petrochemical Spec Library above.

SPI Specification Sheet Tips & Tricks

- Selection of Spec Libraries and Forms need to be based on your project needs. Each user and project is different and no single available Library will have every Spec Forms to fit all projects.
- A modified title box will not show up on the second notes page. An overlaid title box will show on the second sheet. If you modify a title box, create an overlay title box and associate it to sheets that use a second notes page or only use an overlaid title box.



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SPI Specification Sheet Tips & Tricks

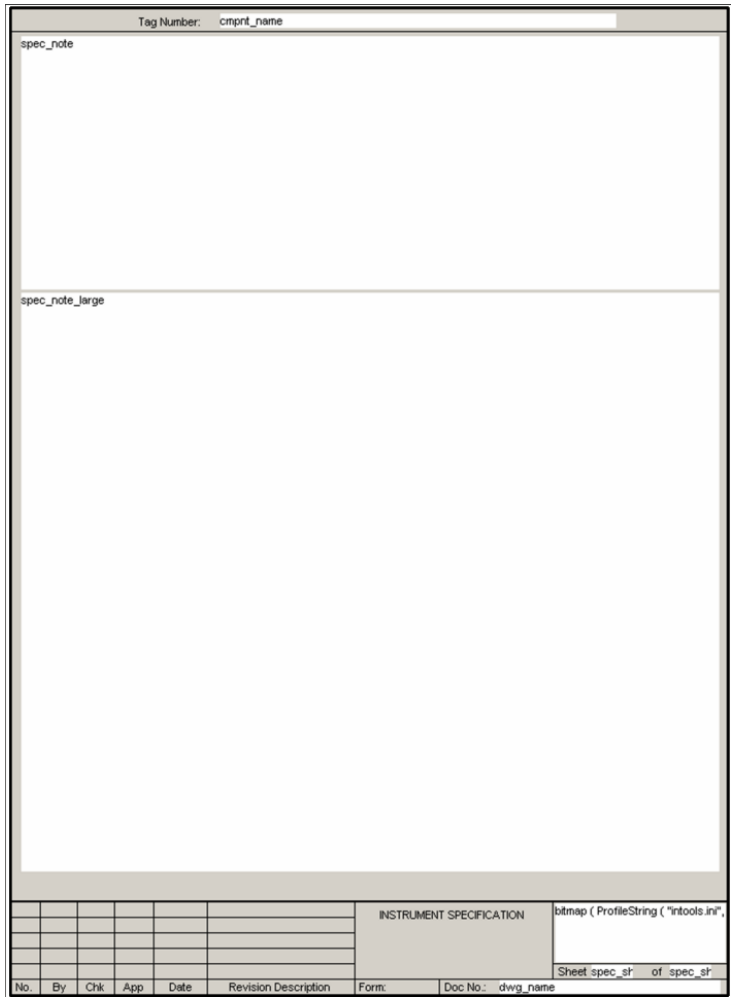
- Always open InfoMaker with the Export icon in SPI to allow InfoMaker to open and link to the project database
- To a provide a margin at the top of datasheet reports, set a parameter “TOPMARGIN = 250” in the INTOOLS.INI file [PRINTER] section.
- When adding a column to a spec page, Use the Page Editor in SPI to select an unused column. InfoMaker allows selection of columns that are already on the Spec.
- Enter the Form Numbers in the Description field of the Page Editor so you know what forms a page is used on.
- When saving Spec Sheet pages to PDF files, Include the Form number in the filename. This makes them easier to recognize when searching for a particular type of Spec.
- Edit Title Block or Overlay Title Block to make ISO 9000 compliant Forms or add fields to make forms project or plant specific

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SPI Specification Sheet Tips & Tricks

- Create your own Notes Page
  - Move notes to Second Page
  - Allows title customizing
  - Add Large Notes
  - Add Process Notes
  - Eliminate the need to “Print Notes on separate sheet”
  - Edit Title Block for ISO-9000 compliant second Page
  - Add your Notes Page to Control Valve and other Spec forms with small Notes area on first page



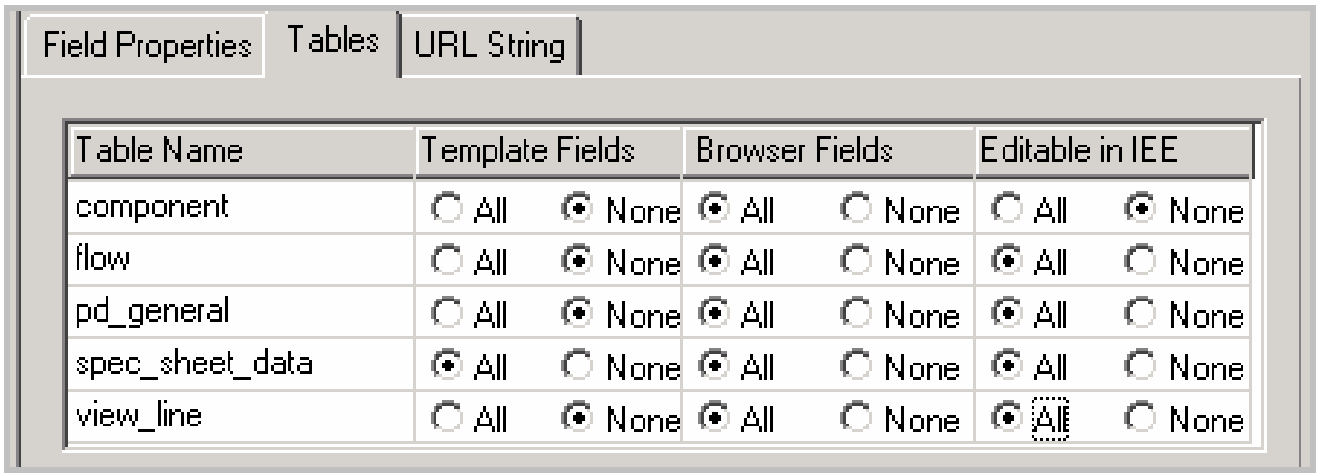
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SPI Specification Sheet Tips & Tricks

Spec Browser Data Dictionary

- SPI Out of the Box Spec Data Dictionary settings are reversed
- In the Spec data dictionary select only the spec_sheet_data for Templates, All tables for Browser and All but component for IEE



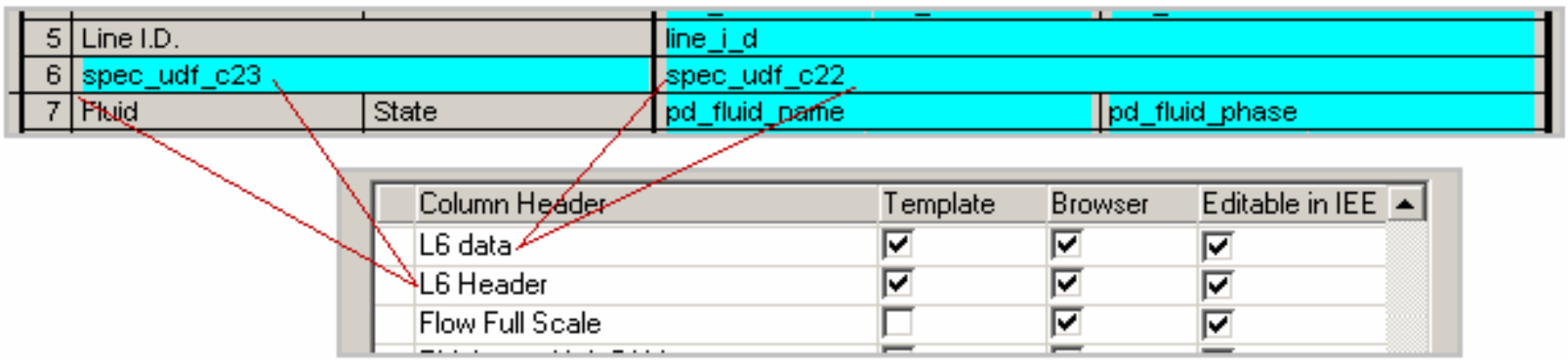
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SPI Specification Sheet Tips & Tricks

Spec Browser Data Dictionary

- Change the headers in the Spec data dictionary to make more sense and shorten to fit the browser area.
- For blank lines on a spec, Name the headers with the line number instead of Spec_udf numbers. This will make more sense in the browser



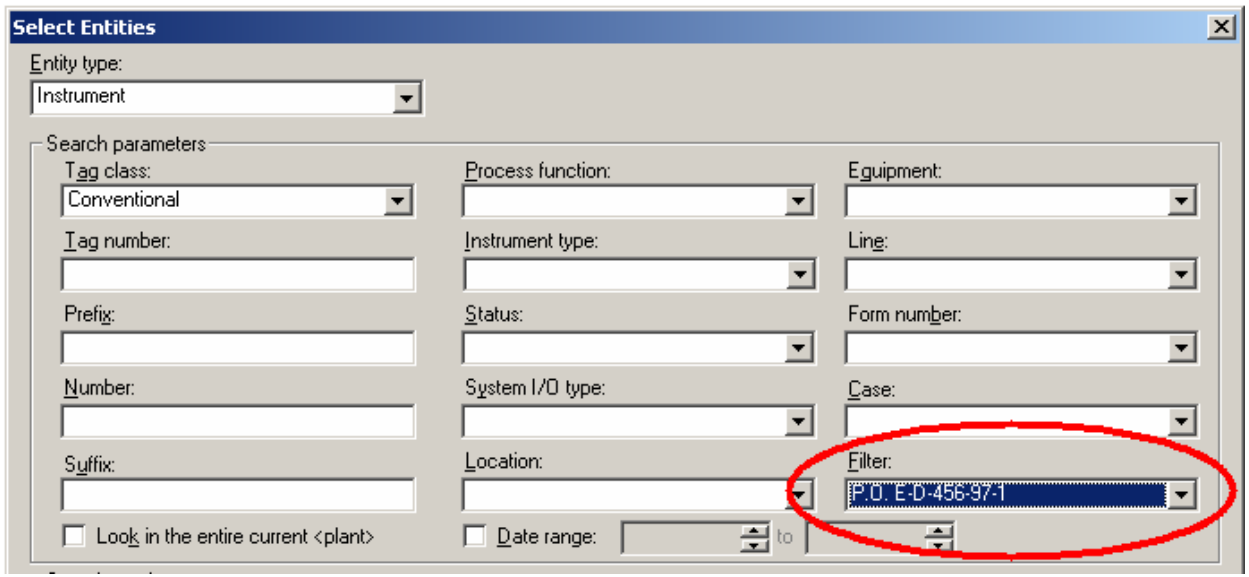
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SPI Specification Sheet Tips & Tricks

Spec Browser Data Dictionary

- To print spec sheets filtered on a specific field:
  - Create and Name a filter in a Spec Browser View
  - Select the Named filter from the “Find” dialog box and select the Specs you wish to print.



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SPI Specification Sheet Libraries

QUESTIONS?

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